INTE 201: PROGRAMMING 3 (Structured Programming)

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Year and Section: DIT 2-3
Date Passed: October 25, 2026

Activities – Week 3 – Control Flow I – Decision-Making
if • if-else • else-if Ladder • Nested-if • switch

BASIC

Simple Grade Classifier

Use a single if-else statement to classify a numeric score as passing or failing.

Requirements:

- Declare an int score variable with a test value.
- Use if-else to print "PASSED" if score is 60 or above, otherwise "FAILED".
- Test your program with at least 2 different score values by changing the variable.

Program Link 1: https://www.jdoodle.com/ia/ah33-cag5-u4gx

Program Link 2: https://www.jdoodle.com/ia/dfq5-wy48-n5fu

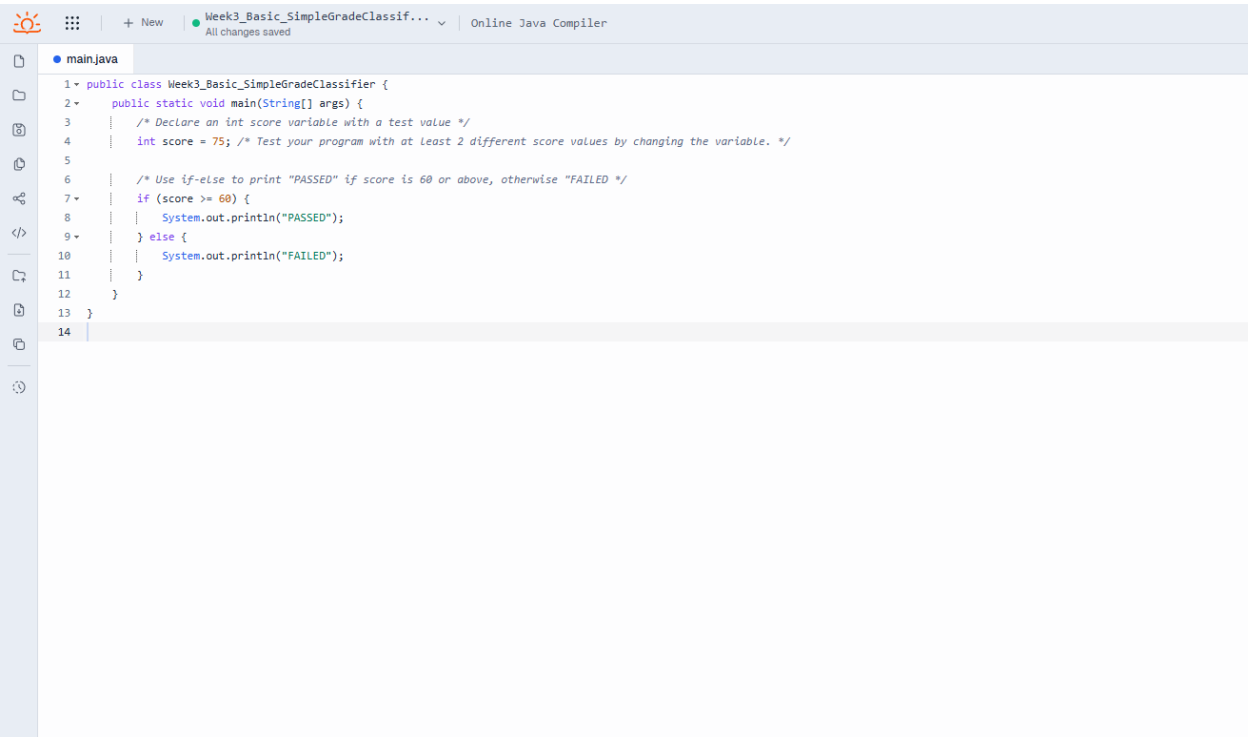
Source Code:

```
public class Week3_Basic_SimpleGradeClassifier {
  public static void main(String[] args) {
    /* Declare an int score variable with a test value */
    int score = 75; /* Test your program with at least 2 different score values by changing the variable. */

    /* Use if-else to print "PASSED" if score is 60 or above, otherwise "FAILED */
    if (score >= 60) {
      System.out.println("PASSED");
    } else {
      System.out.println("FAILED");
    }
  }
}
```

Screenshots:

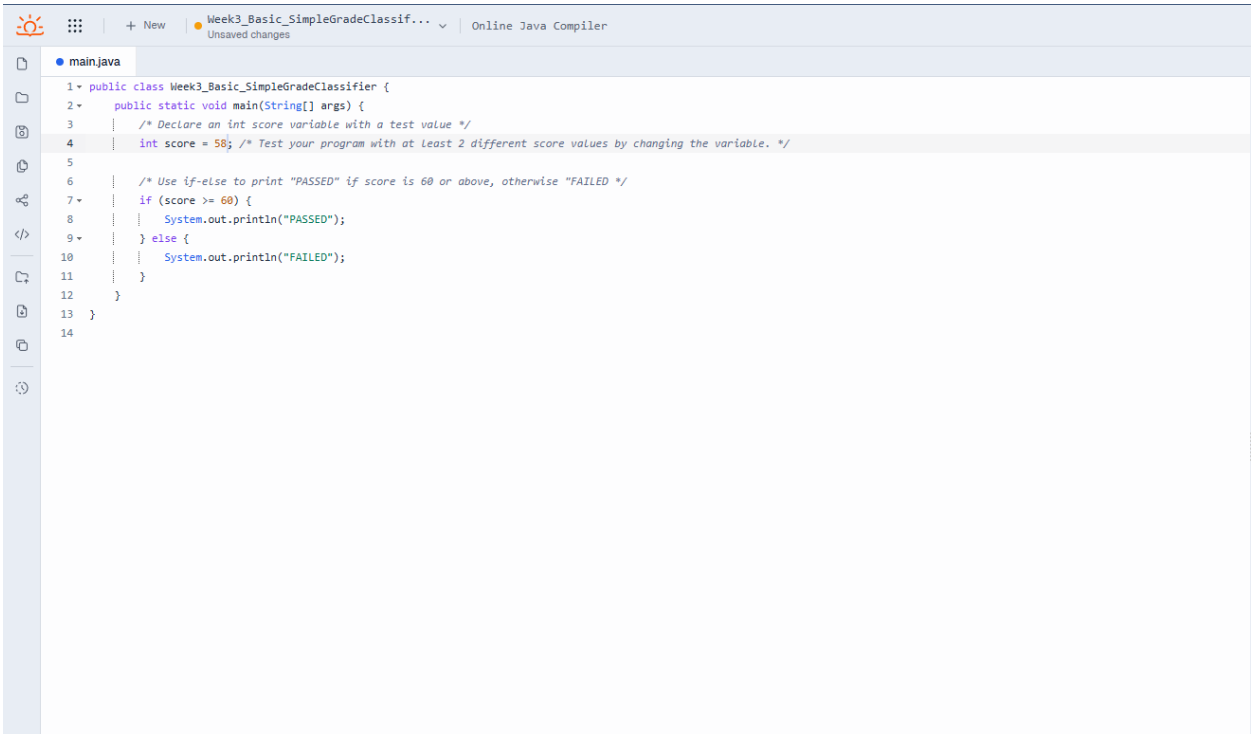
Input (Code):



Output:



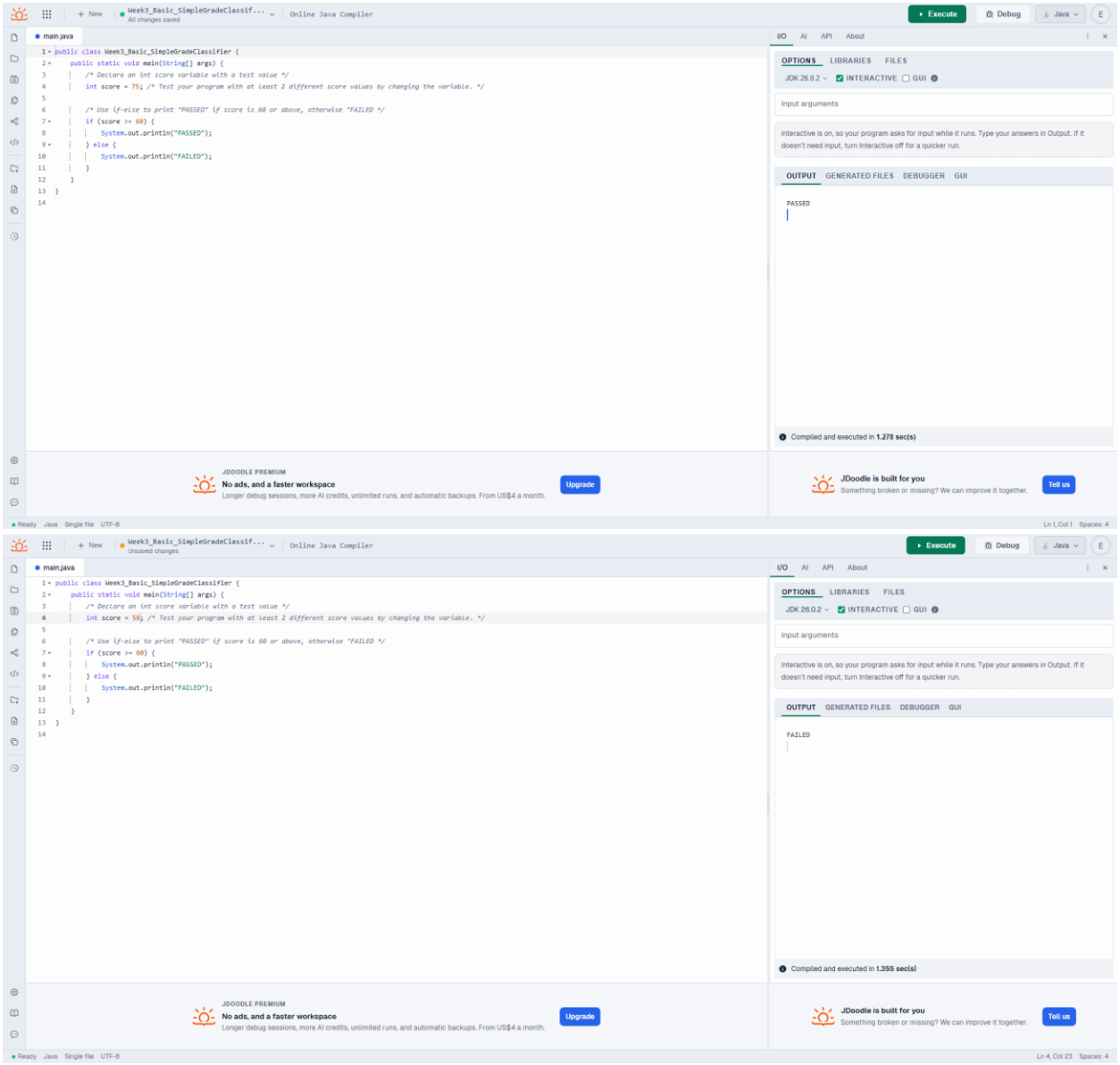
Input (Code):



Output:



Final Screenshot:



Compiler Used:

https://www.jdoodle.com/online-java-compiler

INTERMEDIATE

Multi-Tier Discount System

Use an else-if ladder to assign a discount tier based on purchase amount.

Requirements:

- Declare a double purchaseAmount variable.
- Use an else-if ladder with at least 4 branches to assign a discount percentage (e.g., 0%, 5%, 10%, 15%) based on increasing amount ranges.
- Print both the assigned discount tier and the final discounted price.
- Make sure your ranges are ordered correctly so no tier is unreachable.

Program Link: https://www.jdoodle.com/ia/3yv3-kb37-zuc8

Source Code:

```
public class Week3_Intermediate_MultiTierDiscountSystem {
  public static void main(String[] args) {
    /* Declare a double purchaseAmount variable */
    double purchaseAmount = 150.00;
    double discountPercent = 0.0;
    String tier = "";

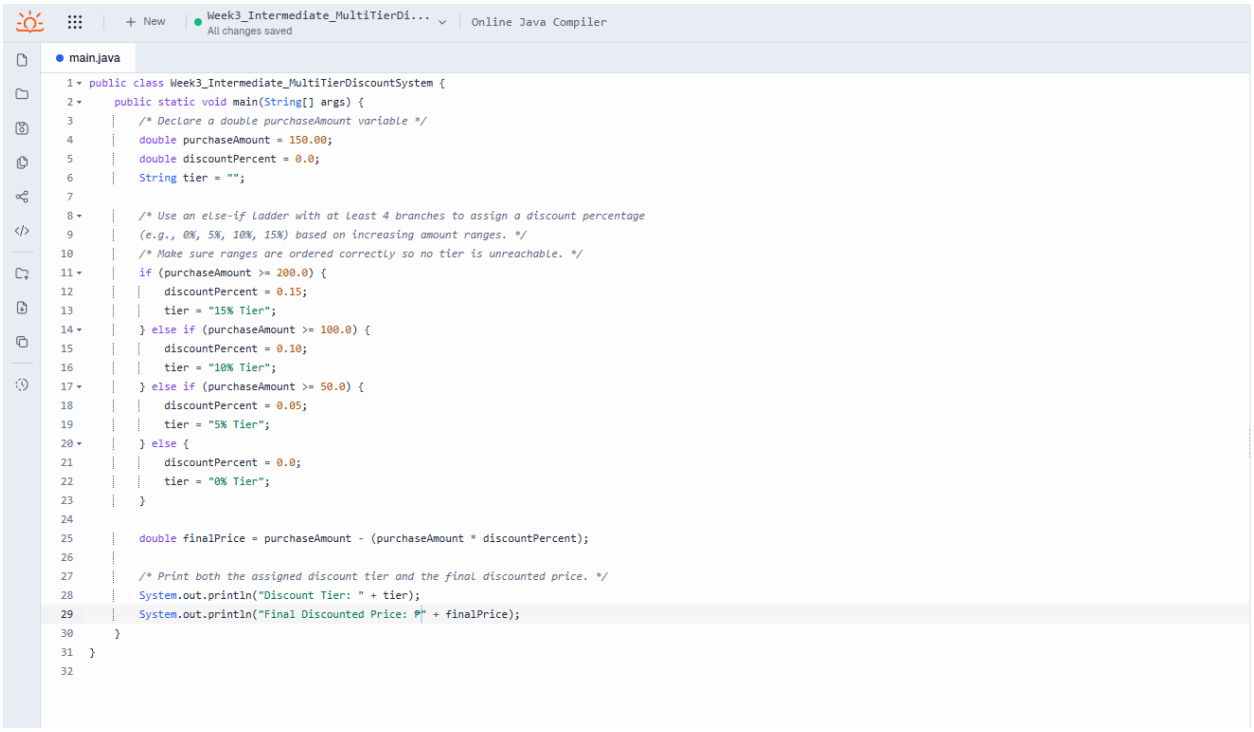
    /* Use an else-if ladder with at least 4 branches to assign a discount percentage
    (e.g., 0%, 5%, 10%, 15%) based on increasing amount ranges. */
    /* Make sure ranges are ordered correctly so no tier is unreachable. */
    if (purchaseAmount >= 200.0) {
      discountPercent = 0.15;
      tier = "15% Tier";
    } else if (purchaseAmount >= 100.0) {
      discountPercent = 0.10;
      tier = "10% Tier";
    } else if (purchaseAmount >= 50.0) {
      discountPercent = 0.05;
      tier = "5% Tier";
    } else {
      discountPercent = 0.0;
      tier = "0% Tier";
    }

    double finalPrice = purchaseAmount - (purchaseAmount * discountPercent);

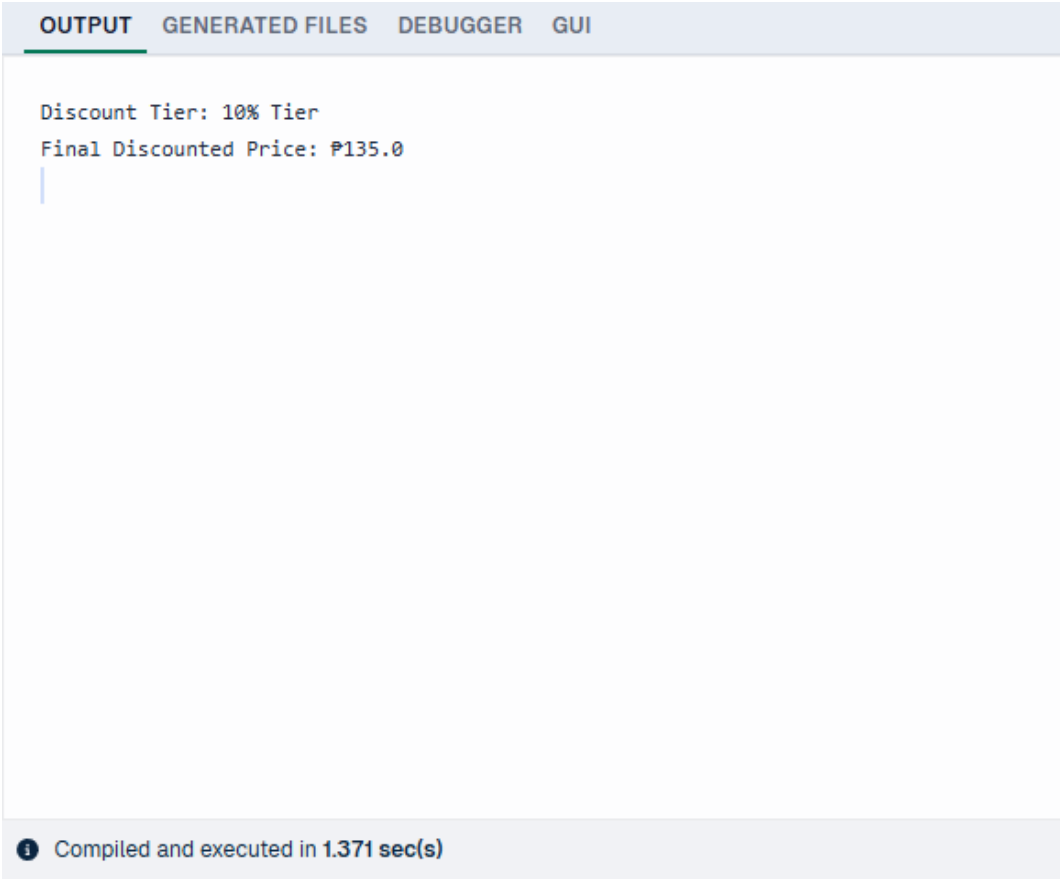
    /* Print both the assigned discount tier and the final discounted price. */
    System.out.println("Discount Tier: " + tier);
    System.out.println("Final Discounted Price: ₱" + finalPrice);
  }
}
```

Screenshots:

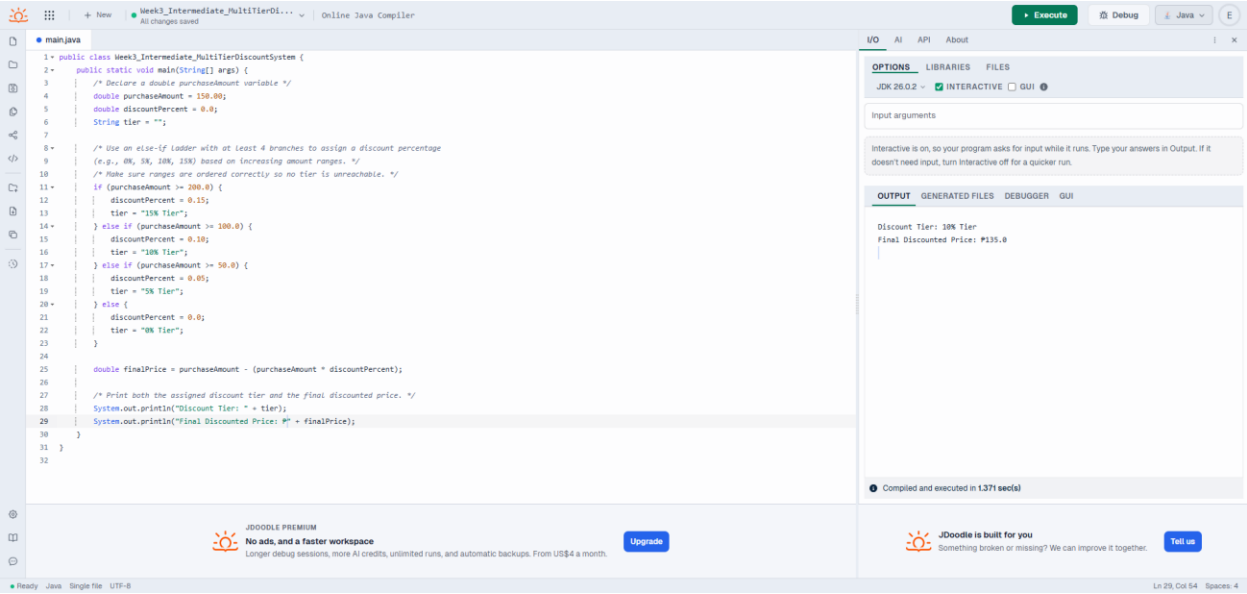
Input (Code):


New
Week3_Intermediate_MultiTierDi...
All changes saved
Online Java Compiler
main.java
public class Week3_Intermediate_MultiTierDiscountSystem {
public static void main(String[] args) {
/* Declare a double purchaseAmount variable */
double purchaseAmount = 150.00;
double discountPercent = 0.0;
String tier = "";
/* Use an else-if ladder with at least 4 branches to assign a discount percentage
(e.g., 0%, 5%, 10%, 15%) based on increasing amount ranges. */
/* Make sure ranges are ordered correctly so no tier is unreachable. */
if (purchaseAmount >= 200.0) {
discountPercent = 0.15;
tier = "15% Tier";
} else if (purchaseAmount >= 100.0) {
discountPercent = 0.10;
tier = "10% Tier";
} else if (purchaseAmount >= 50.0) {
discountPercent = 0.05;
tier = "5% Tier";
} else {
discountPercent = 0.0;
tier = "0% Tier";
}
double finalPrice = purchaseAmount - (purchaseAmount * discountPercent);
/* Print both the assigned discount tier and the final discounted price. */
System.out.println("Discount Tier: " + tier);
System.out.println("Final Discounted Price: ₱" + finalPrice);
}
}

Output:


OUTPUT
GENERATED FILES
DEBUGGER
GUI
Discount Tier: 10% Tier
Final Discounted Price: ₱135.0
Compiled and executed in 1.371 sec(s)

Final Screenshot:



Compiler Used:

https://www.jdoodle.com/online-java-compiler

ADVANCED

Nested Login Validator

Use nested if statements to validate a login in stages, only checking each stage if the previous one passed.

Requirements:

- Check the username first; only if it matches, check the password inside a nested if.
- Only if BOTH match, check a third condition inside an even deeper nested if (e.g., account is not locked).

- Print a specific, different failure message depending on which stage failed.
- Test all failure paths by deliberately providing wrong values at each stage.

Program Link: https://www.jdoodle.com/ia/5wqj-39sm-ft6p

Source Code:

```
public class Week3_Advanced_NestedLoginValidator {
  public static void main(String[] args) {
    String dbUser = "admin";
    String dbPass = "secure123";
    boolean isAccountLocked = false;

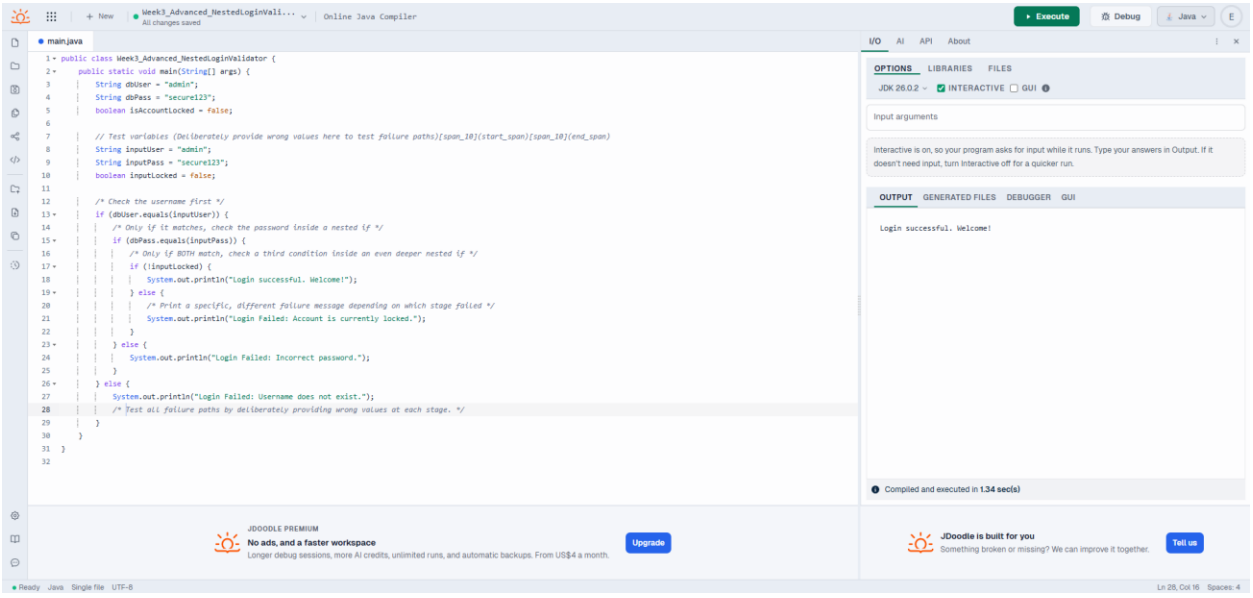
    // Test variables (Deliberately provide wrong values here to test failure paths)[span_10](start_span)[span_10](end_span)
    String inputUser = "admin";
    String inputPass = "secure123";
    boolean inputLocked = false;

    /* Check the username first */
    if (dbUser.equals(inputUser)) {
      /* Only if it matches, check the password inside a nested if */
      if (dbPass.equals(inputPass)) {
        /* Only if BOTH match, check a third condition inside an even deeper nested if */
        if (!inputLocked) {
          System.out.println("Login successful. Welcome!");
        } else {
          /* Print a specific, different failure message depending on which stage failed */
          System.out.println("Login Failed: Account is currently locked.");
        }
      } else {
        System.out.println("Login Failed: Incorrect password.");
      }
    } else {
      System.out.println("Login Failed: Username does not exist.");
      /* Test all failure paths by deliberately providing wrong values at each stage. */
    }
  }
}
```

Screenshots:

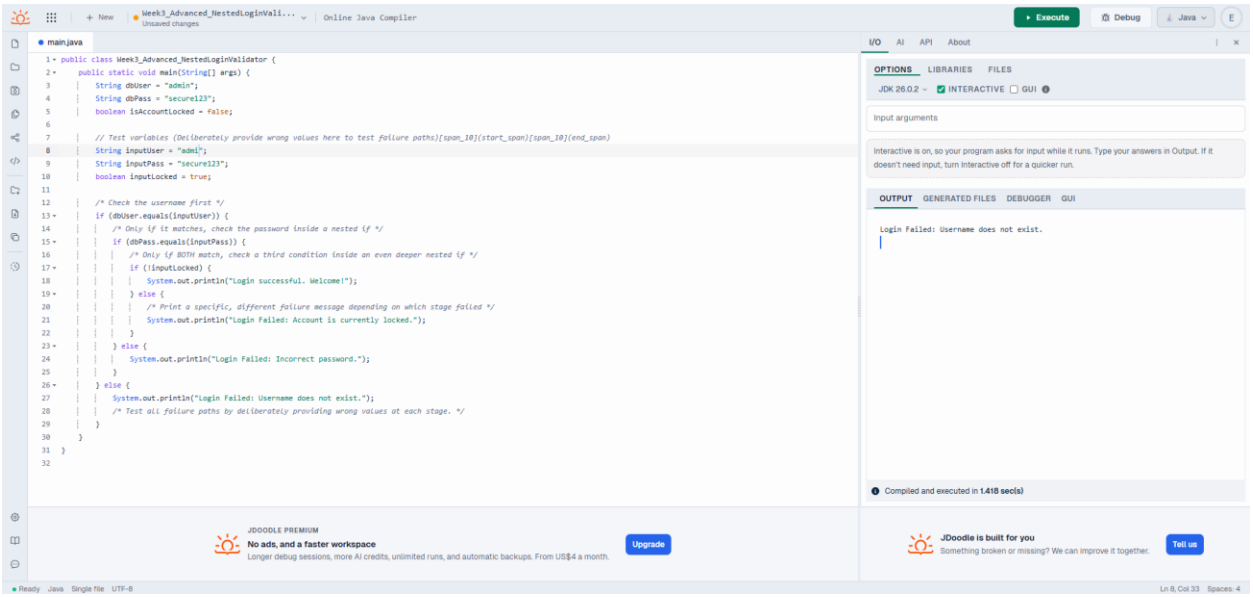
Result 1: Login Successful

Program Link: https://www.jdoodle.com/ia/5wqj-39sm-ft6p



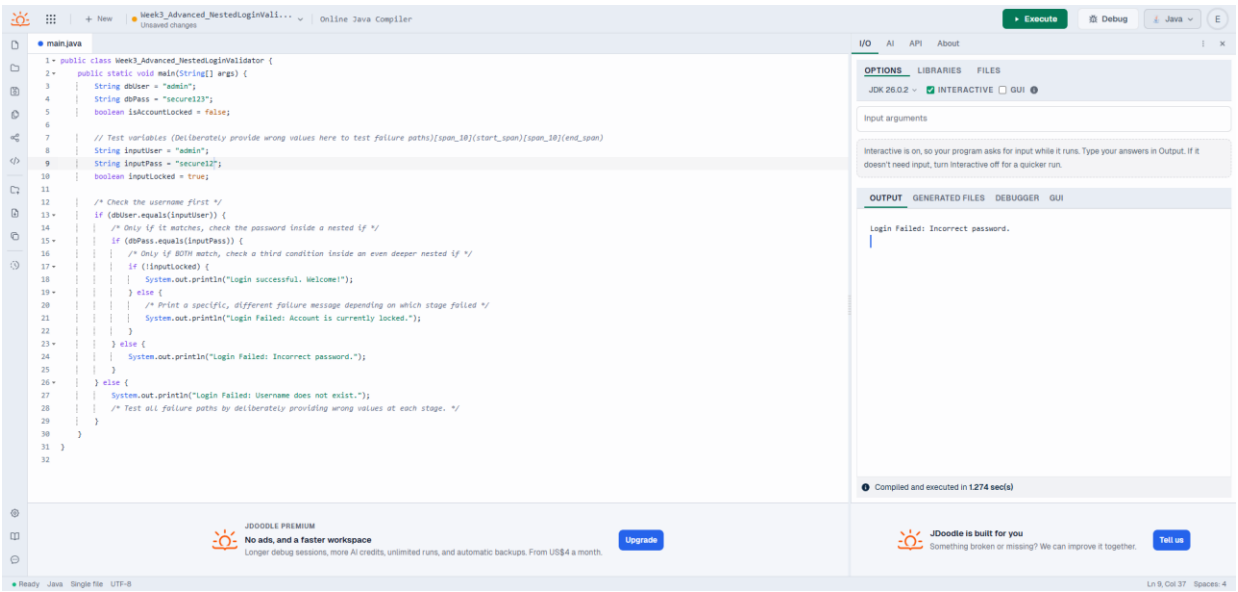
Result 2: Login Failed - Username does not exist

Program Link: https://www.jdoodle.com/ia/a7hz-7zkj-ht95



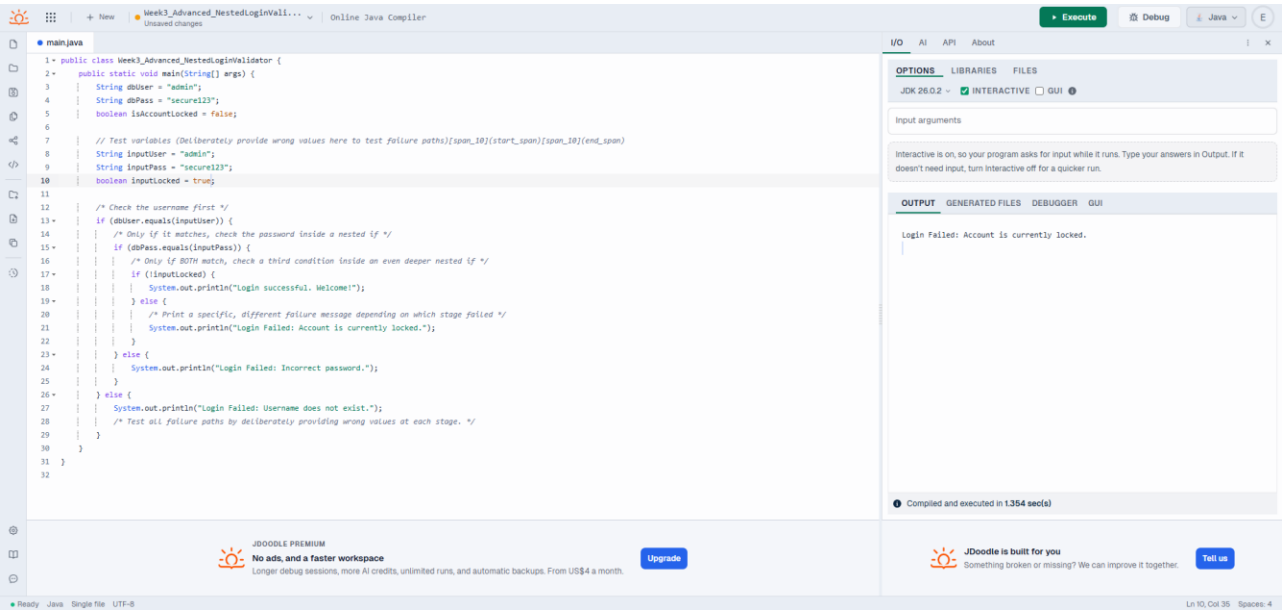
Result 3: Login Failed – Incorrect Password

Program Link: https://www.jdoodle.com/ia/3mjd-gu3x-h9p6



Result 4: Login Failed – Account is Currently Locked

Program Link: https://www.jdoodle.com/ia/m2fr-p89c-du86



Compiler Used:

https://www.jdoodle.com/online-java-compiler

EXPERT

ATM Transaction Authorizer

Build a complete authorization system combining every selection statement from this week - no loops allowed.

Requirements:

- Use a nested if to check a username AND a 4-digit PIN before allowing anything else to proceed.
- Use an else-if ladder to determine the account's tier (Bronze/Silver/Gold/Platinum) from its balance.
- Use a nested if in the withdrawal logic to check sufficient balance AND a tier-dependent daily limit.
- Use a switch on account tier to select a transaction fee percentage, then combine everything with if-else to APPROVE or DENY, printing the exact reason for any denial.

Program Link: https://www.jdoodle.com/ia/ed49-4ear-vu4s

Source Code:

```
public class Week3_Expert_ATMTransactionAuthorizer {
  public static void main(String[] args) {
    String dbUser = "ethangueta";
    int dbPin = 4568;
    double balance = 6500.0;

    String inputUser = "ethangueta";
    int inputPin = 4568;
    double withdrawalAmount = 2500.0;

    /* Use a nested if to check a username AND a 4-digit PIN before allowing anything else to proceed. */
    if (dbUser.equals(inputUser)) {
      if (dbPin == inputPin) {

        String tier = "";
        double dailyLimit = 0.0;

        /*    Use an else-if ladder to determine the account's tier (Bronze/Silver/Gold/Platinum) from its balance. */
        if (balance >= 10000) { tier = "Platinum"; dailyLimit = 5000; }
        else if (balance >= 5000) { tier = "Gold"; dailyLimit = 3000; }
        else if (balance >= 1000) { tier = "Silver"; dailyLimit = 1000; }
        else { tier = "Bronze"; dailyLimit = 500; }

        /*    Use a nested if in the withdrawal logic to check sufficient balance AND a tier-dependent daily limit. */
        double feeRate;
        switch (tier) {
          case "Platinum": feeRate = 0.0; break;
          case "Gold": feeRate = 0.01; break;
          case "Silver": feeRate = 0.02; break;
          case "Bronze": feeRate = 0.05; break;
          default: feeRate = 0.05; break;
        }

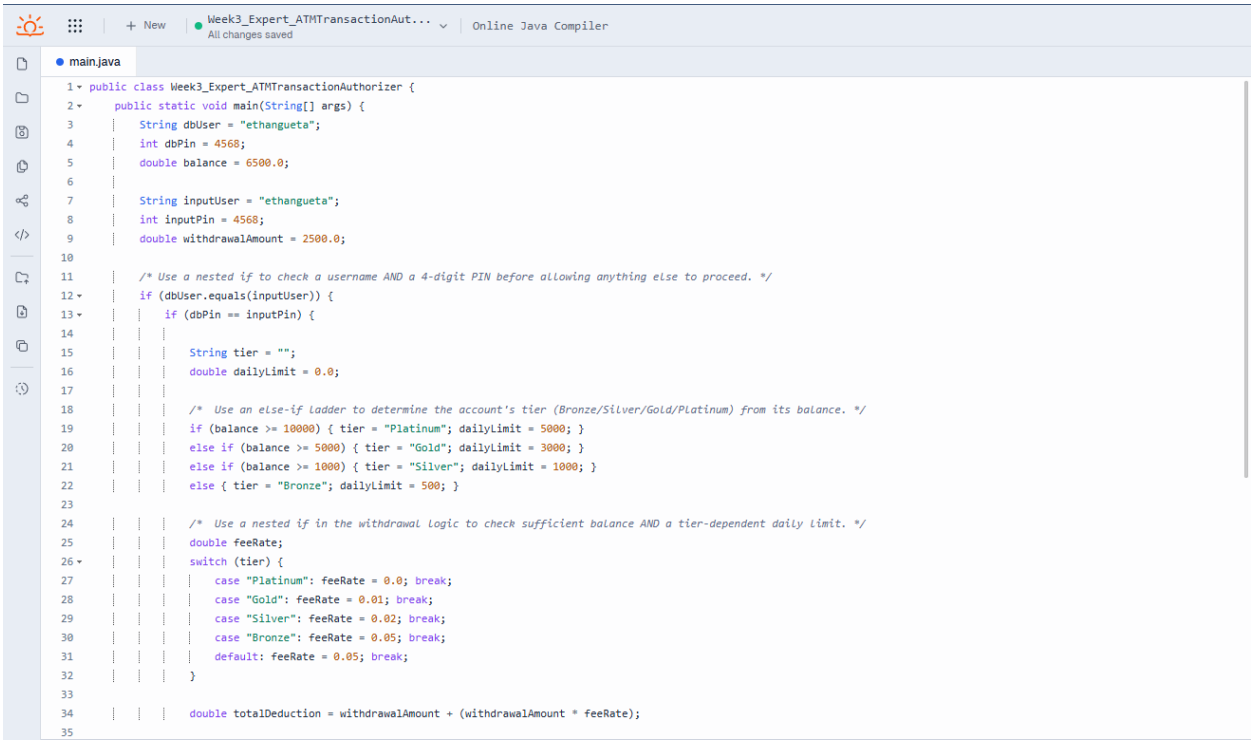
        double totalDeduction = withdrawalAmount + (withdrawalAmount * feeRate);

        /* • Use a switch on account tier to select a transaction fee percentage, then combine everything with
        if-else to APPROVE or DENY, printing the exact reason for any denial.*/
        if (withdrawalAmount <= dailyLimit) {
          if (totalDeduction <= balance) {
            System.out.println("APPROVE: Transaction successful.");
            System.out.println("Remaining Balance: ₱" + (balance - totalDeduction));
          } else {
            System.out.println("DENY: Insufficient balance to cover withdrawal and " + (feeRate * 100) + "% fee.");
          }
        } else {
          System.out.println("DENY: Withdrawal exceeds the " + tier + " tier daily limit of ₱" + dailyLimit);
        }

      } else {
        System.out.println("DENY: Incorrect 4-digit PIN.");
      }
    } else {
      System.out.println("DENY: Invalid username.");
    }
  }
}
```

Screenshots:

Input (Code):




```java
public class Week3_Expert_ATMTransactionAuthorizer {
    public static void main(String[] args) {
        String dbUser = "ethangueta";
        int dbPin = 4568;
        double balance = 6500.0;

        String inputUser = "ethangueta";
        int inputPin = 4568;
        double withdrawalAmount = 2500.0;

        /* Use a nested if to check a username AND a 4-digit PIN before allowing anything else to proceed. */
        if (dbUser.equals(inputUser)) {
            if (dbPin == inputPin) {

                String tier = "";
                double dailyLimit = 0.0;

                /*  Use an else-if ladder to determine the account's tier (Bronze/Silver/Gold/Platinum) from its balance. */
                if (balance >= 10000) { tier = "Platinum"; dailyLimit = 5000; }
                else if (balance >= 5000) { tier = "Gold"; dailyLimit = 3000; }
                else if (balance >= 1000) { tier = "Silver"; dailyLimit = 1000; }
                else { tier = "Bronze"; dailyLimit = 500; }

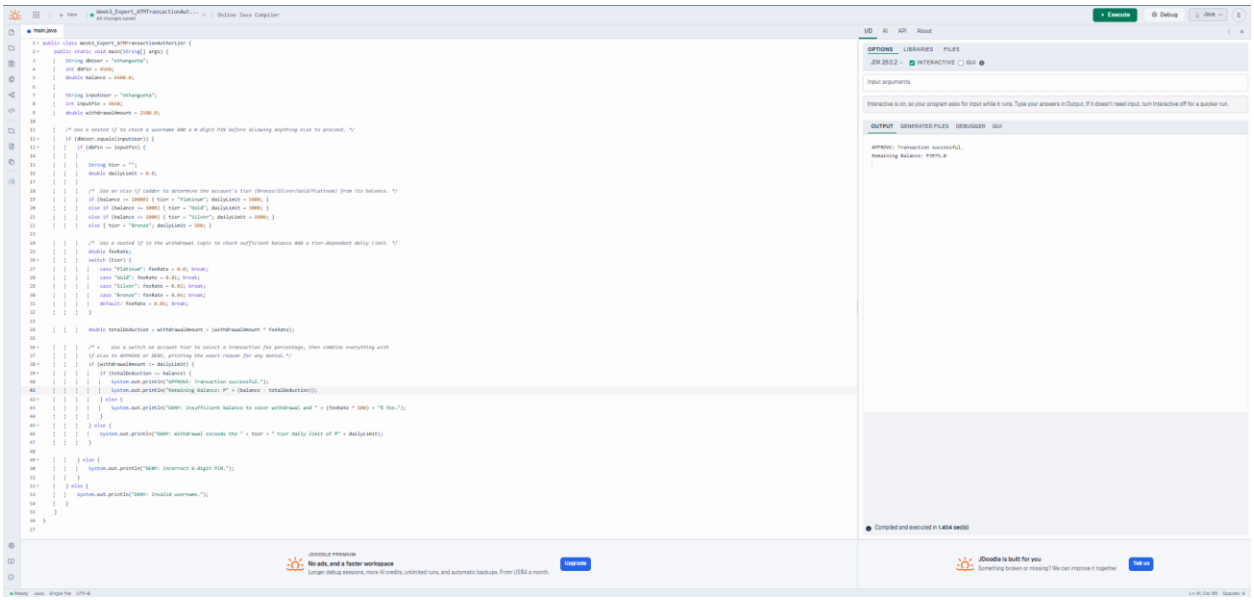
                /*  Use a nested if in the withdrawal logic to check sufficient balance AND a tier-dependent daily limit. */
                double feeRate;
                switch (tier) {
                    case "Platinum": feeRate = 0.0; break;
                    case "Gold": feeRate = 0.01; break;
                    case "Silver": feeRate = 0.02; break;
                    case "Bronze": feeRate = 0.05; break;
                    default: feeRate = 0.05; break;
                }

                double totalDeduction = withdrawalAmount + (withdrawalAmount * feeRate);
```

Output:



Final Screenshot:



Compiler Used:

https://www.jdoodle.com/online-java-compiler